

Learning Cluttered Scene Manipulation with Goals

Lirui Wang, June 9th, 2020

A senior thesis presentation in partial fulfillment of the requirements for the degree of
Bachelor of Science With Departmental Honors



Manipulation in Industry



Quote

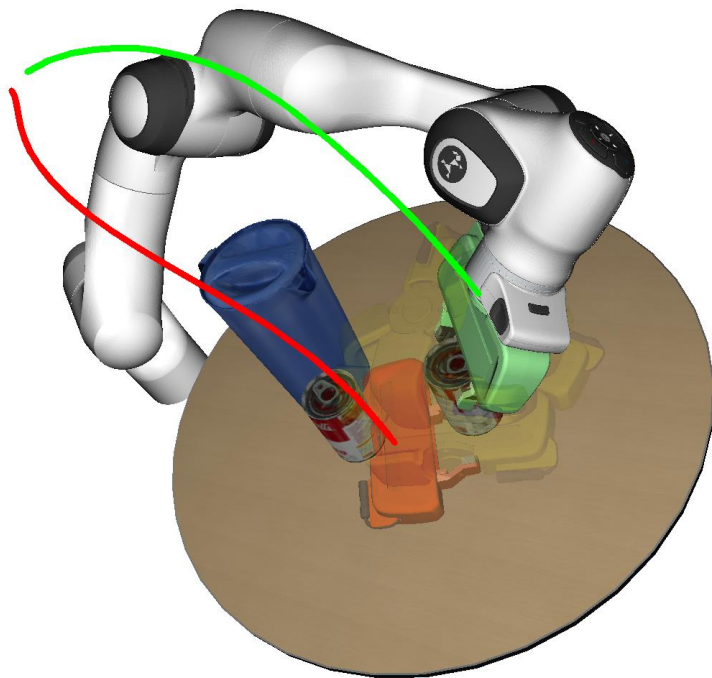
Based on a survey over eleven thousand controllers in the refining, chemicals and pulp and paper industries, 97% of regulatory controllers utilize a PID feedback control algorithm.

-- L.Desborough and R. Miller, 2002

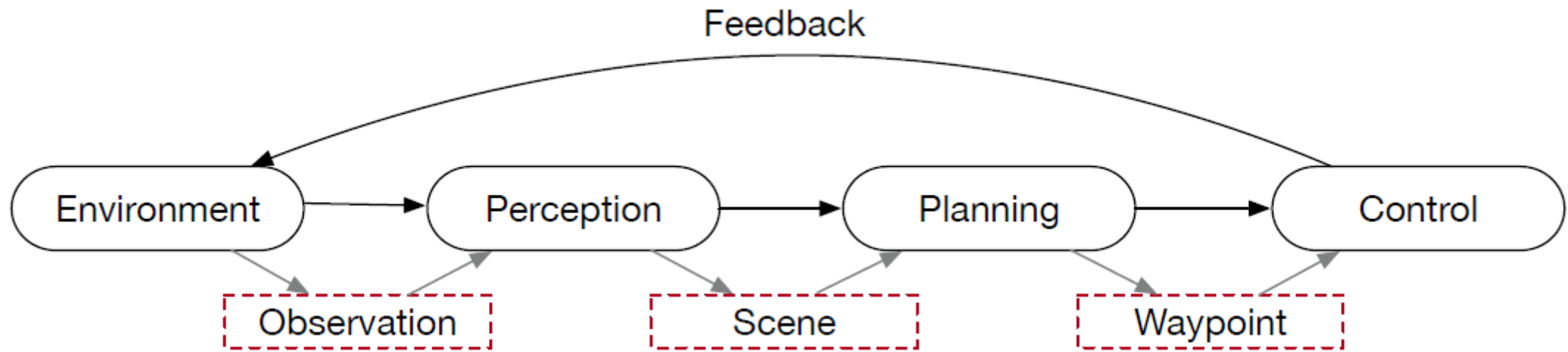
Like a human, our agents construct and learn their own knowledge directly from raw inputs, such as vision, without any hand-engineered features or domain heuristics.

-- D.Silver, 2016

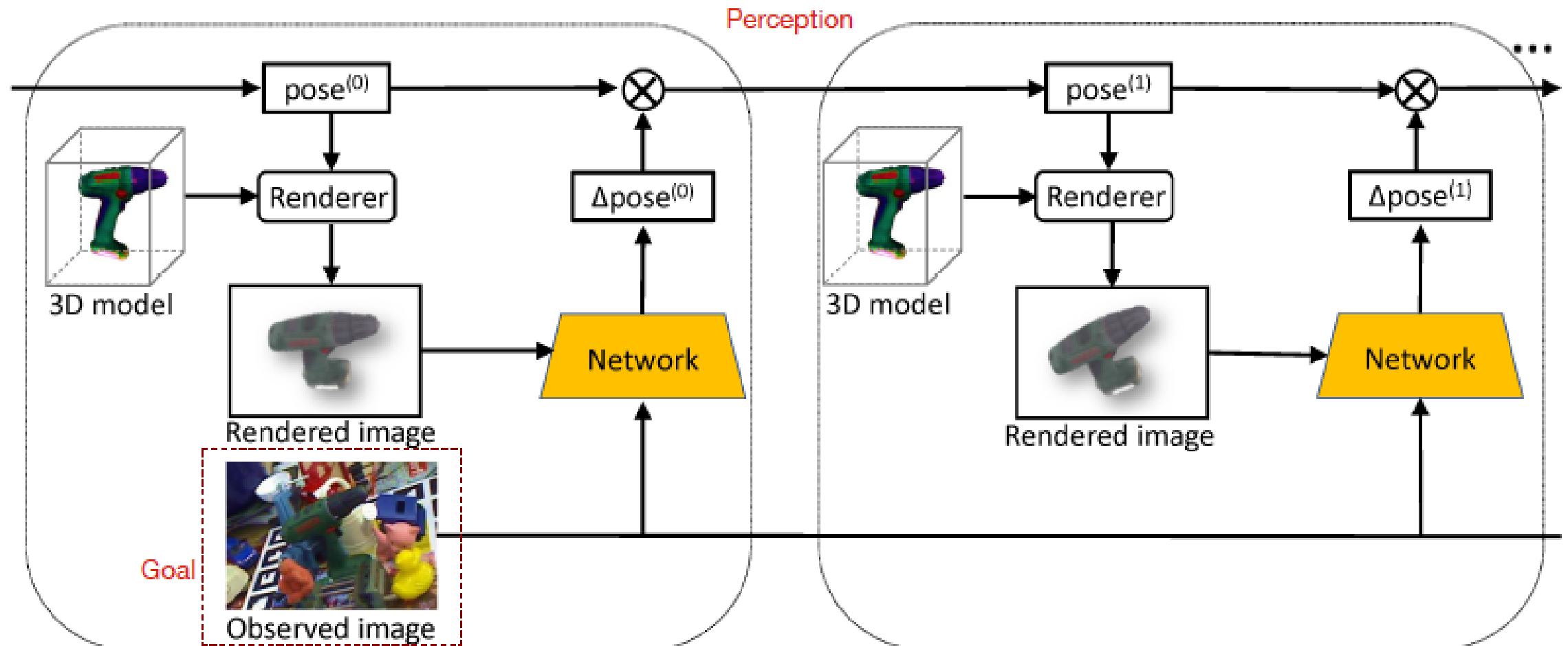
Manipulation in Research



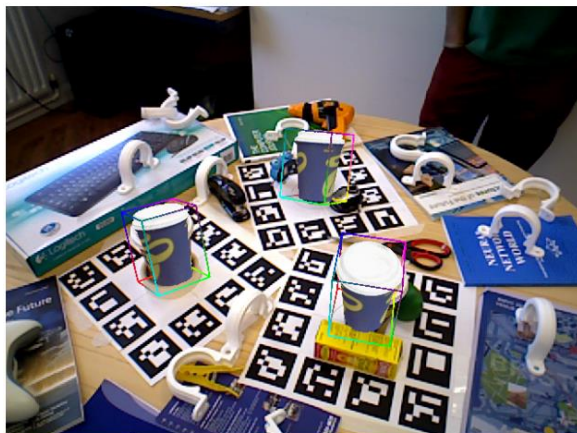
Goals play a role



Perception



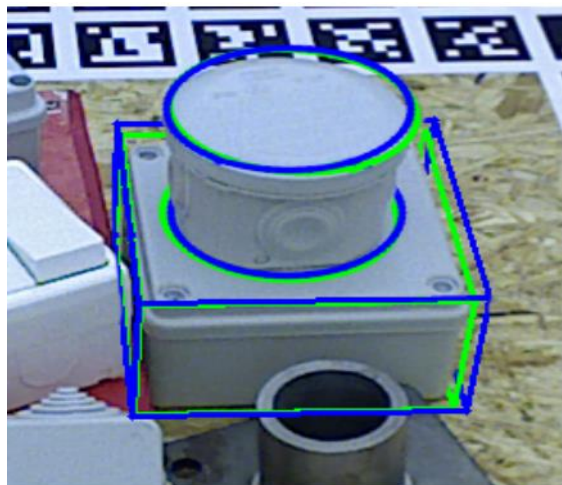
Pose Estimation and Refinement



Kehl et al., ECCV, 2016



Xiang et al., RSS, 2018



Rad et al., ICCV, 2017

RGB

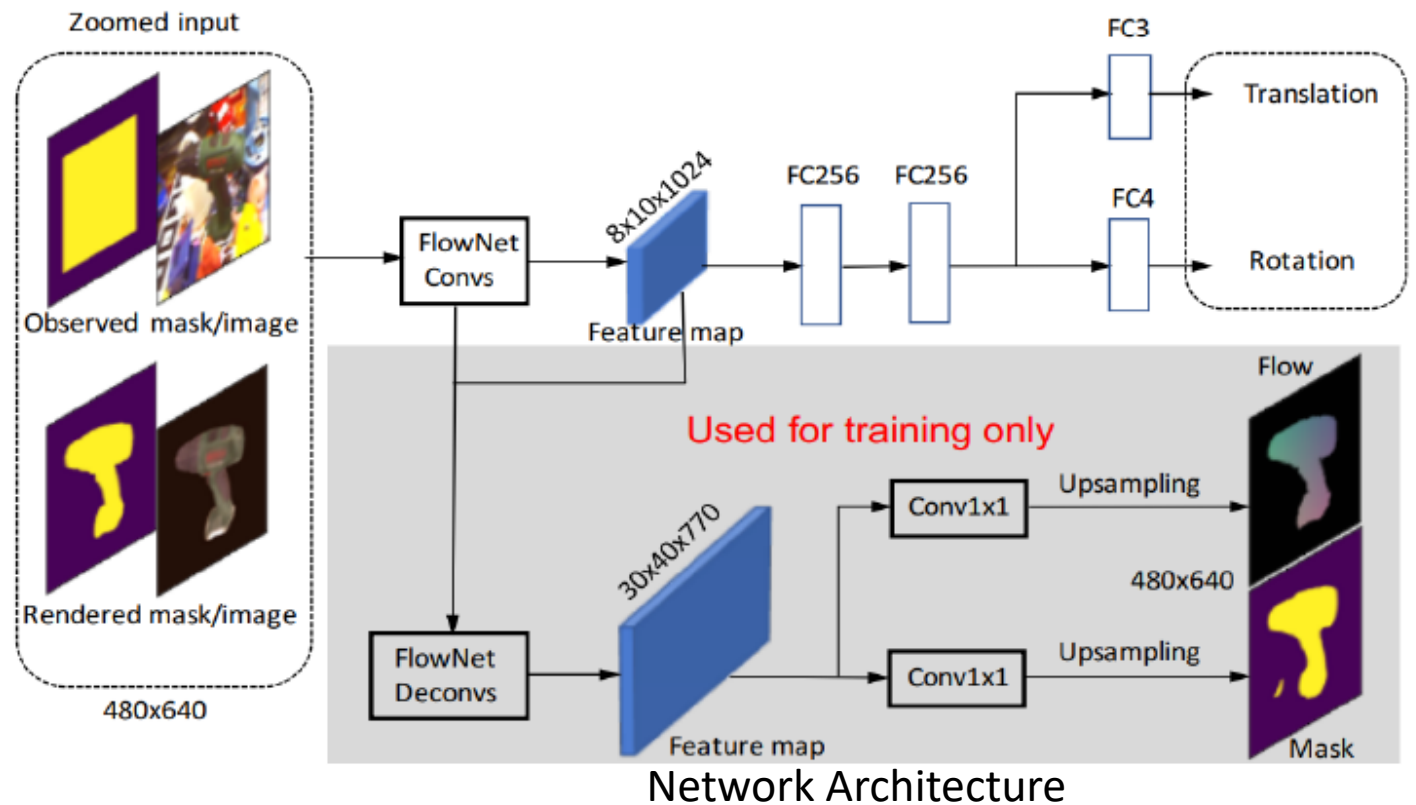


Wang et al., CVPR, 2019

RGBD

DeepIM Recap

- Zoom-In
- Untangled Representation
- Iterative Refinement
- Natural Loss
- Tracking and Application



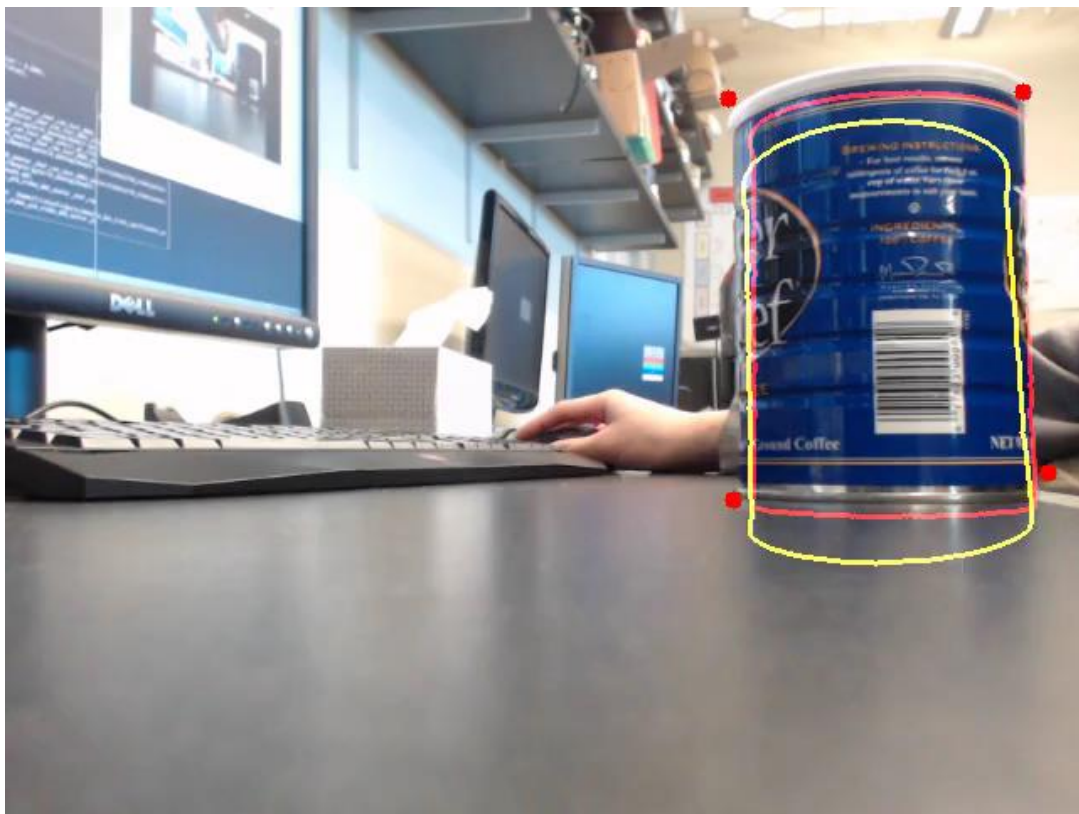
Synthetic Training



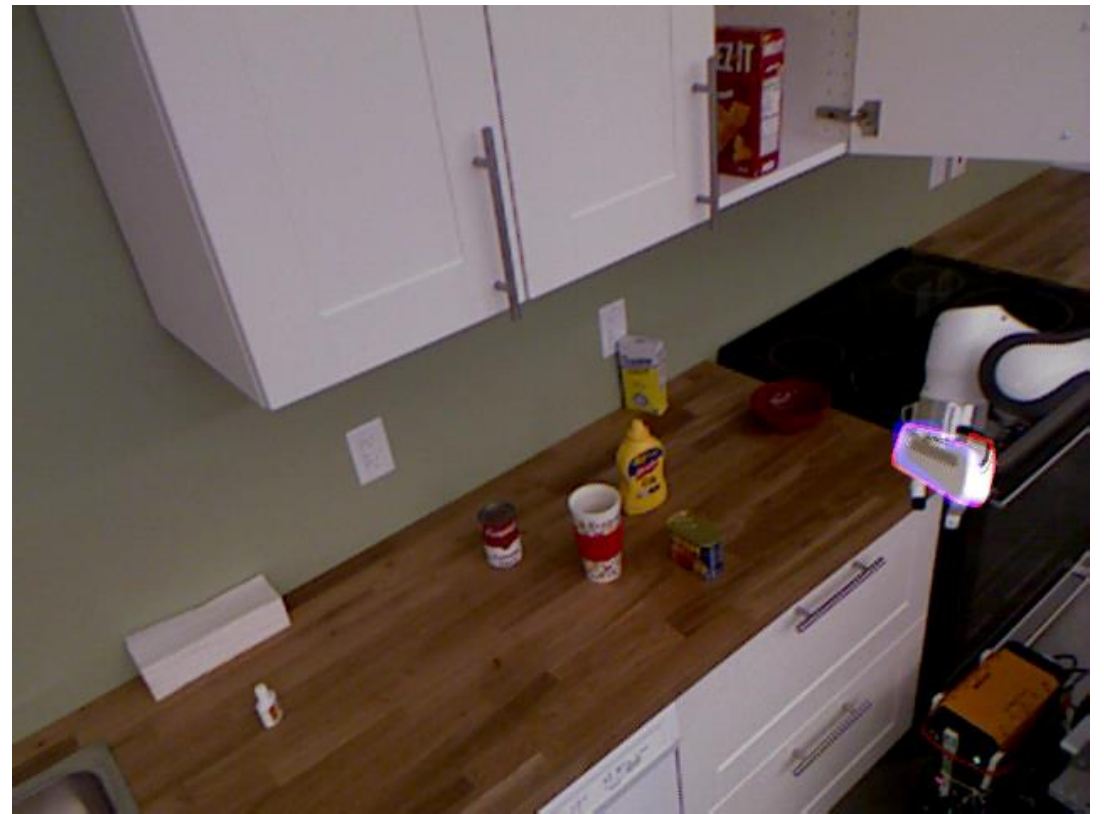
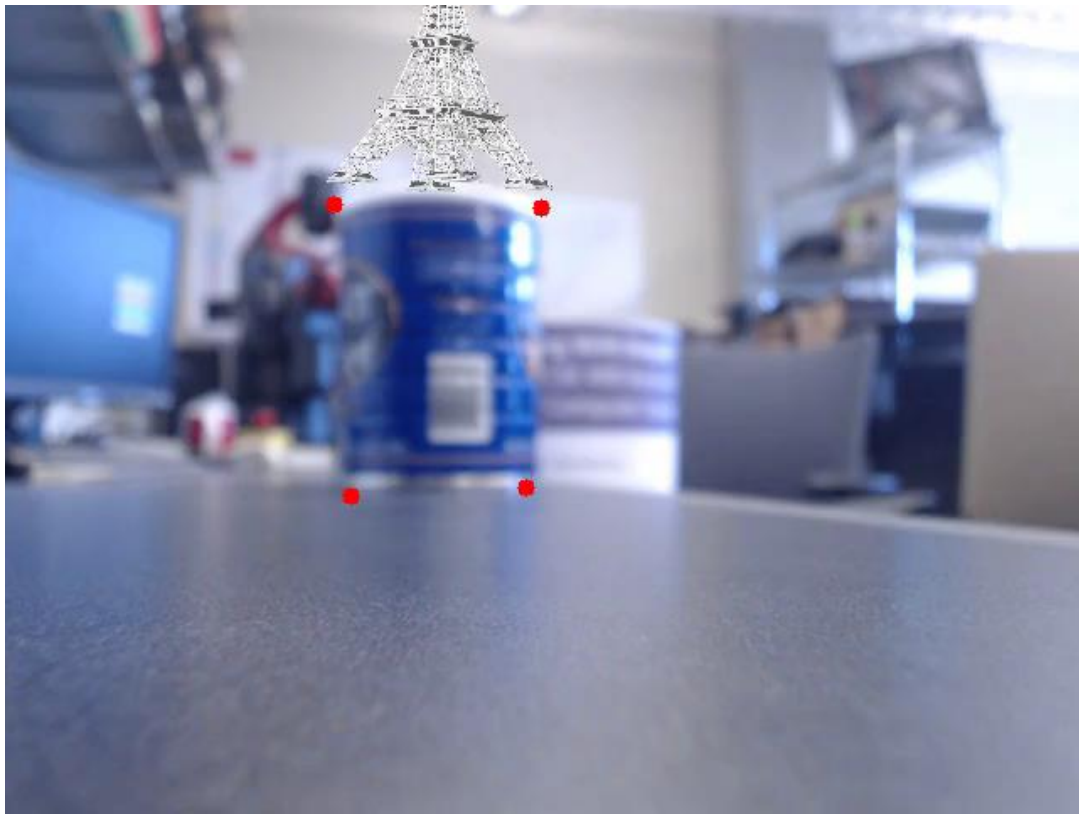
Tracking Performance

Method	Tracking			Reinitialize			PoseCNN		
Metric	$5^\circ, 5cm$	Pose	Projection	$5^\circ, 5cm$	Pose	Projection	$5^\circ, 5cm$	Pose	Projection
Master Chef Can	0.369	0.771	0.231	0.379	0.750	0.254	0.042	0.102	0.001
Sugar Box	0.564	0.745	0.441	0.418	0.581	0.258	0.1362	0.443	0.071
Potted Meat Can	0.402	0.866	0.606	0.375	0.813	0.588	0.106	0.427	0.418
Mustard Bottle	0.799	0.983	0.647	0.811	0.971	0.658	0.193	0.764	0.064
Power Drill	0.371	0.786	0.094	0.276	0.756	0.085	0.089	0.420	0.033
Cracker Box	0.453	0.812	0.241	0.391	0.498	0.074	0.027	0.284	0.001
Mean	0.493	0.827	0.376	0.391	0.441	0.319	0.098	0.406	0.098

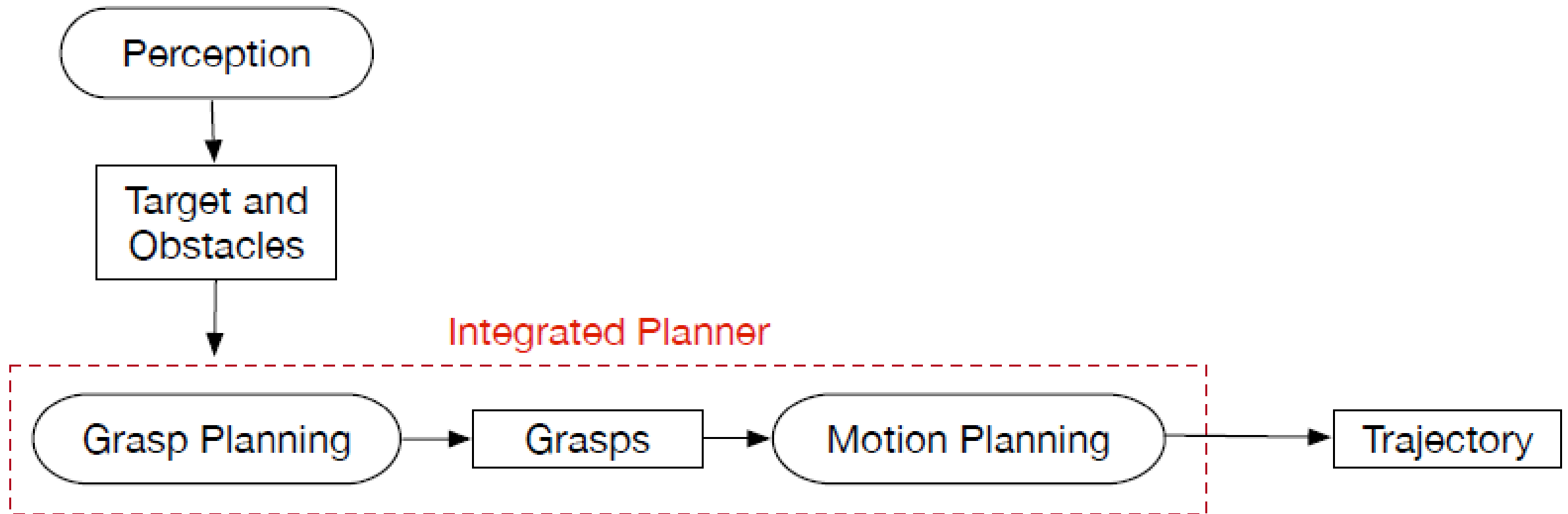
Pose Tracking



Application

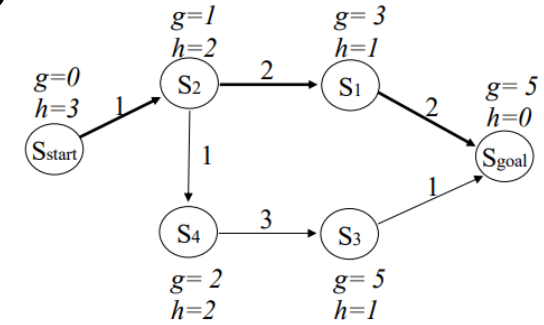


Planning

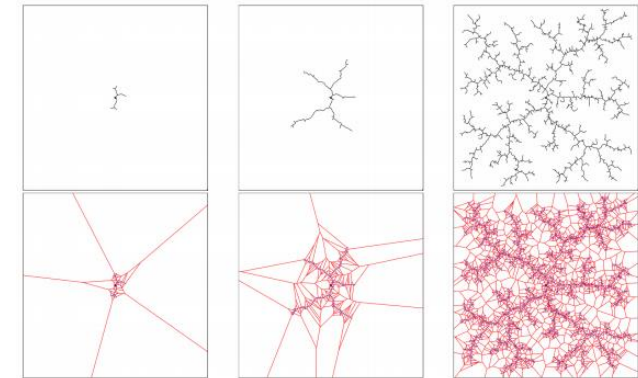


Motion Planning

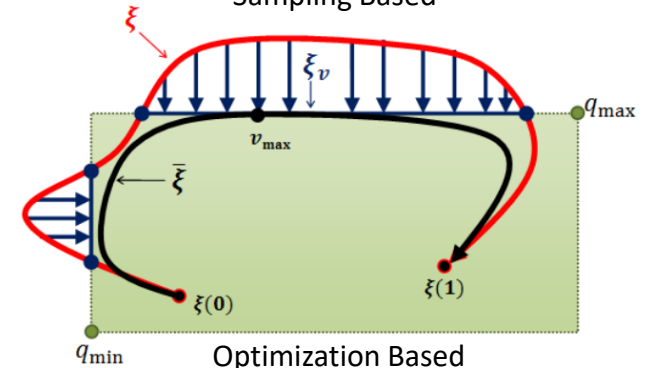
- A* (Deterministic Search)
- PRM/RRT (Sampling)
- CHOMP (Gradient)
- STOMP (Path Integral)
- TrajOpt (SQP)
- GPMP2 (Inference)
- ...



Search Based



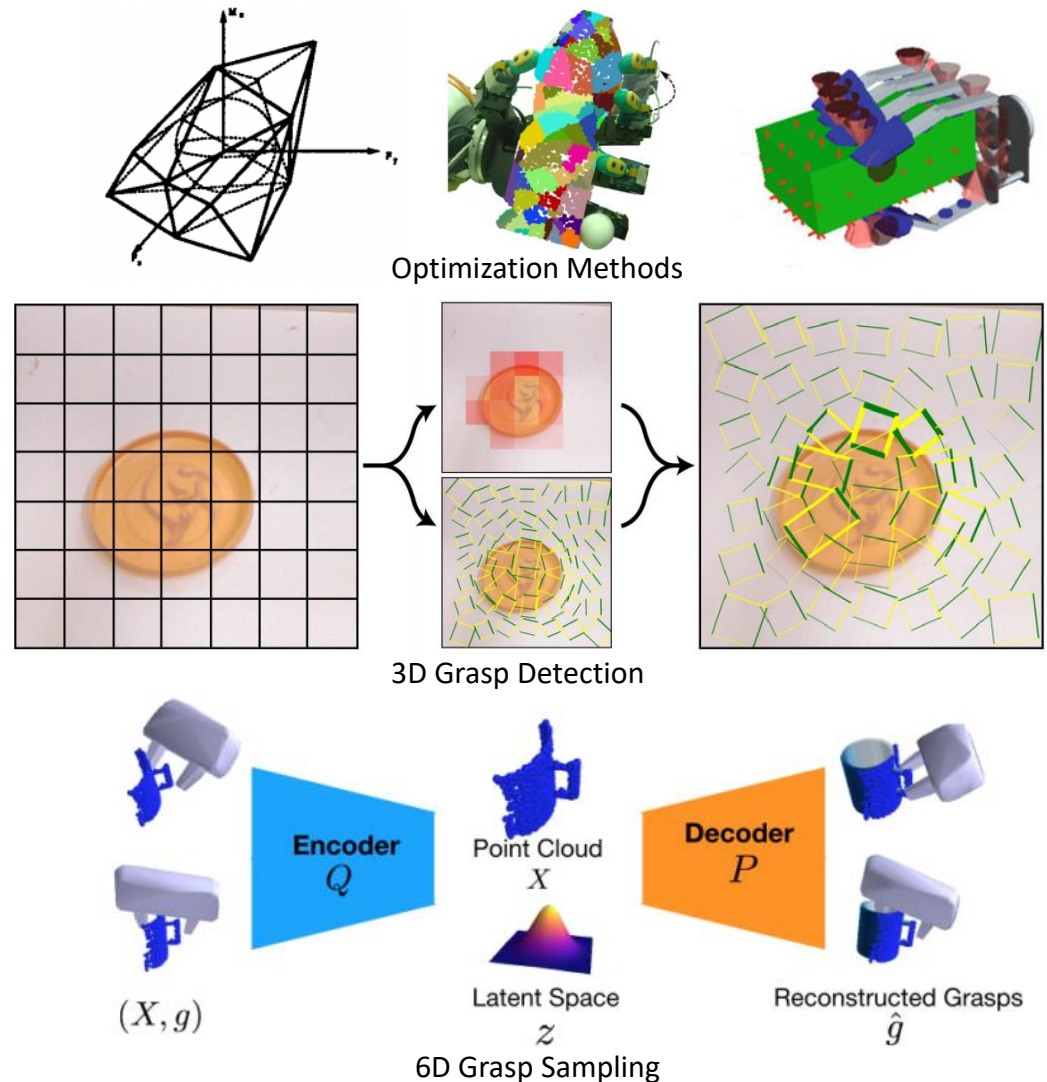
Sampling Based



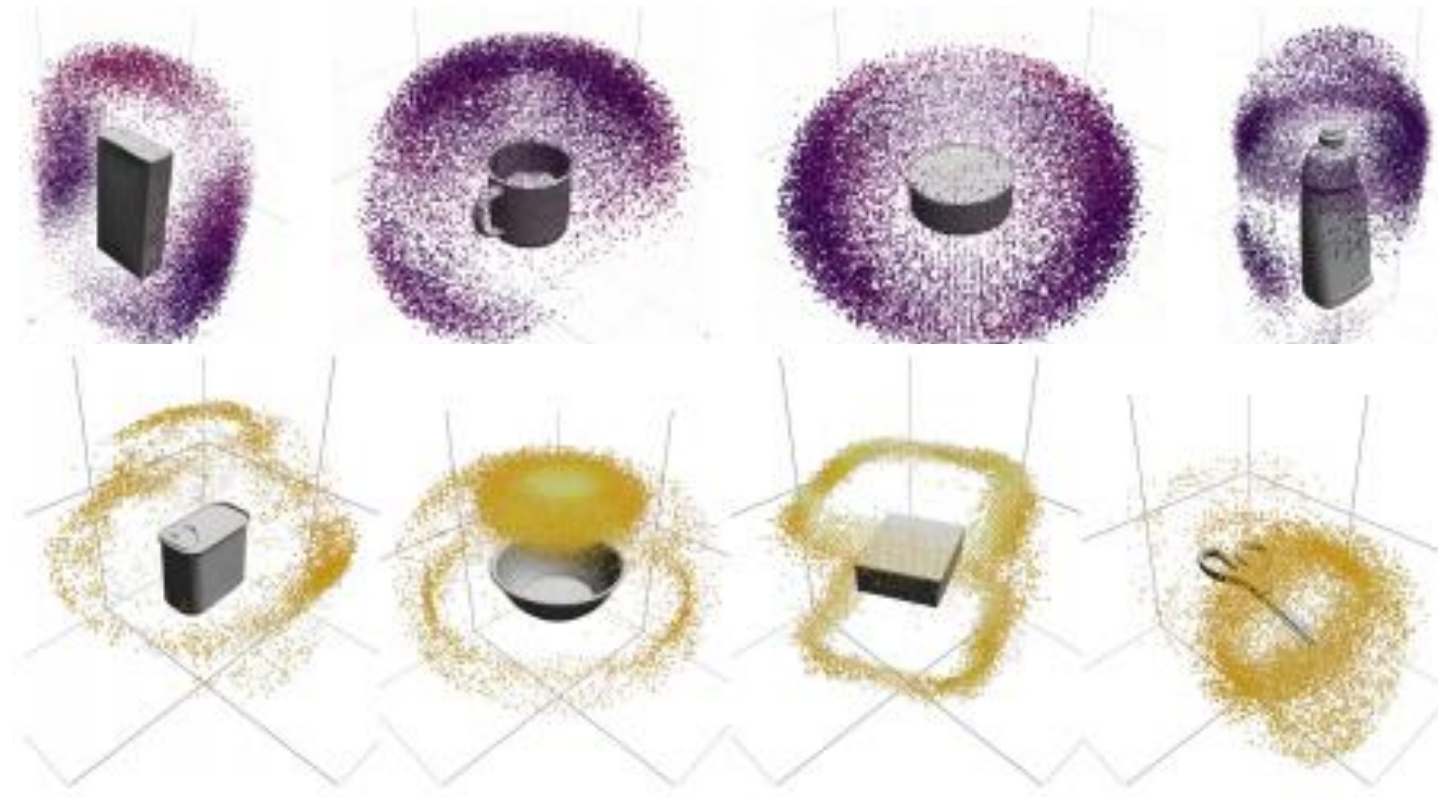
Optimization Based

Grasp Planning

- Analytic Approach
 - Grasp Quality Measure
 - Optimization
- Data-Driven Approach
 - Discriminative Approach
 - Generative Approach

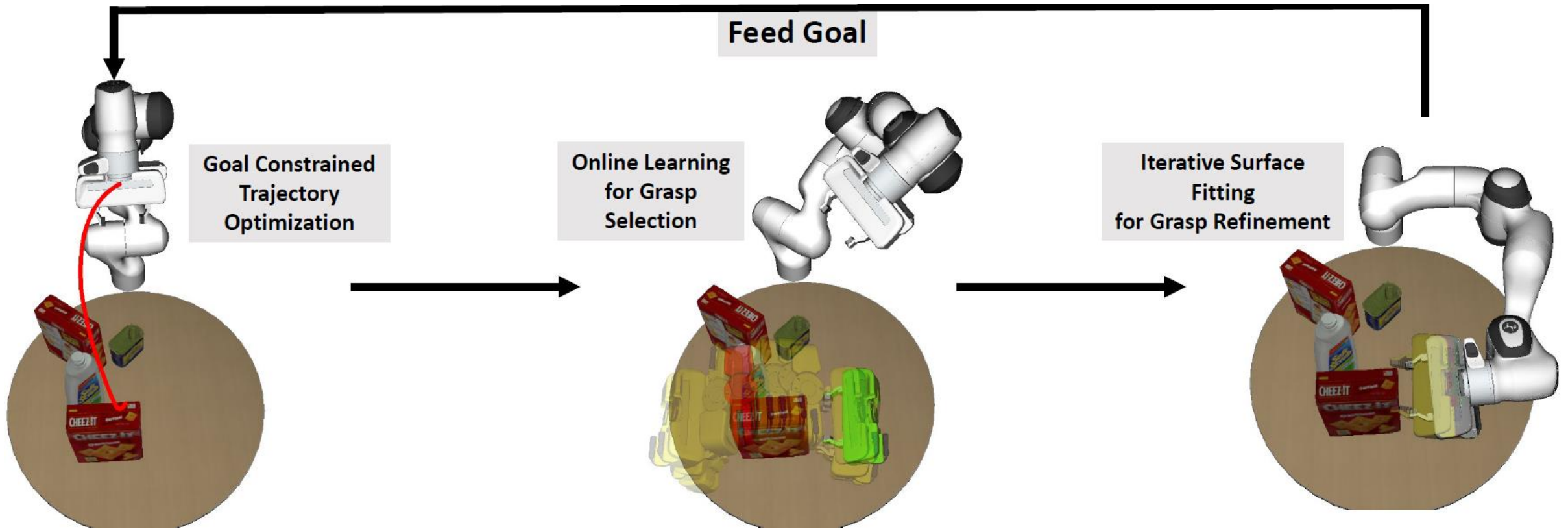


We can have many grasps



$$\xi^* = \arg \min_{\xi} f_{\text{motion}}(\xi) \\ \text{s.t. } \xi(1) \in \mathcal{G}$$

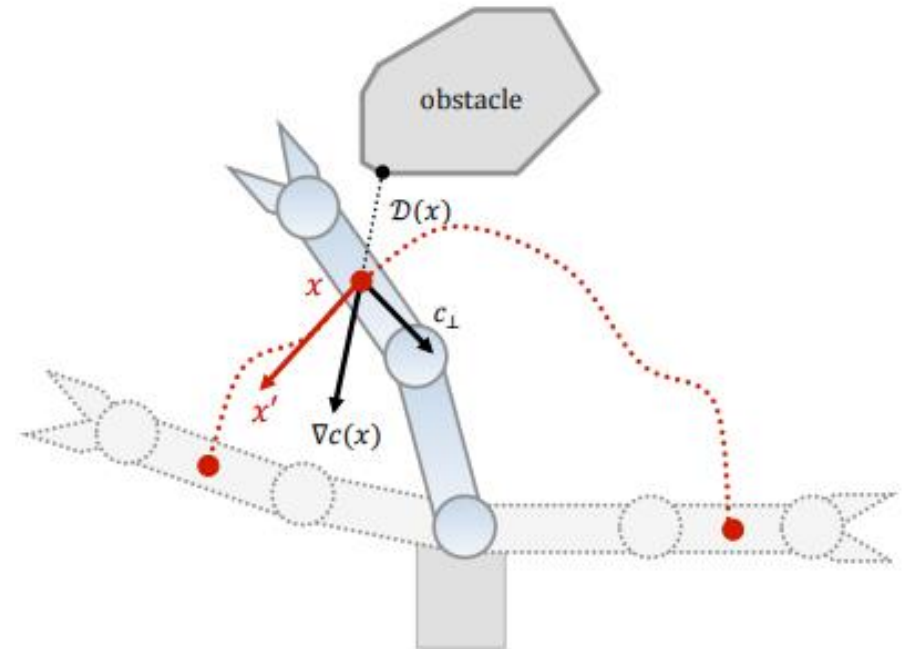
System Overview



CHOMP Recap

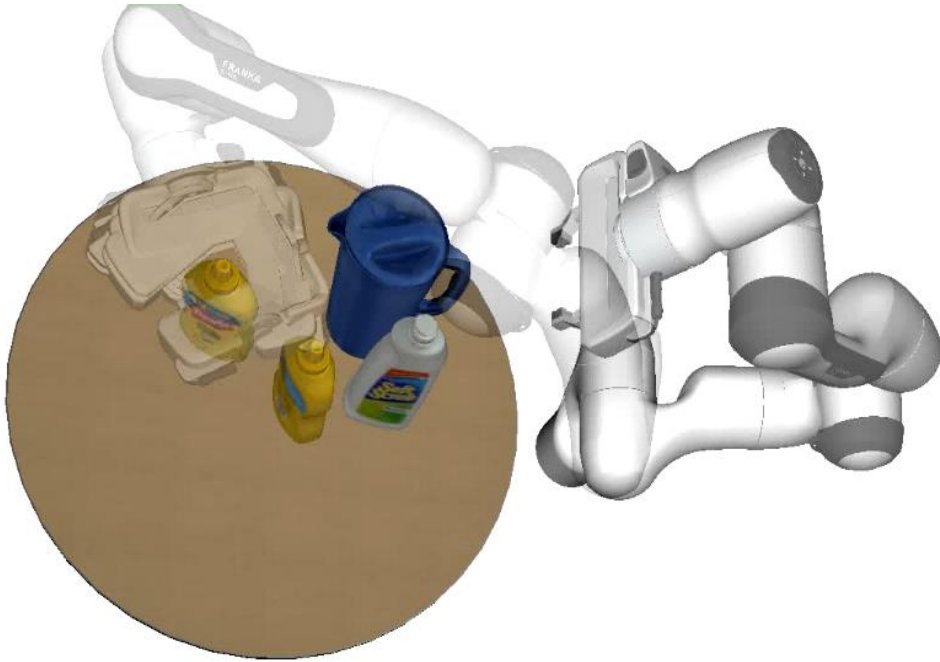
- Covariant Gradient Descent with Smoothness Metric
- Avoid Obstacles with Functional Gradient
- Constraint Satisfaction with Lagrangian

$$f_{\text{motion}}(\xi) = f_{\text{obstacle}}(\xi) + \lambda f_{\text{prior}}(\xi)$$

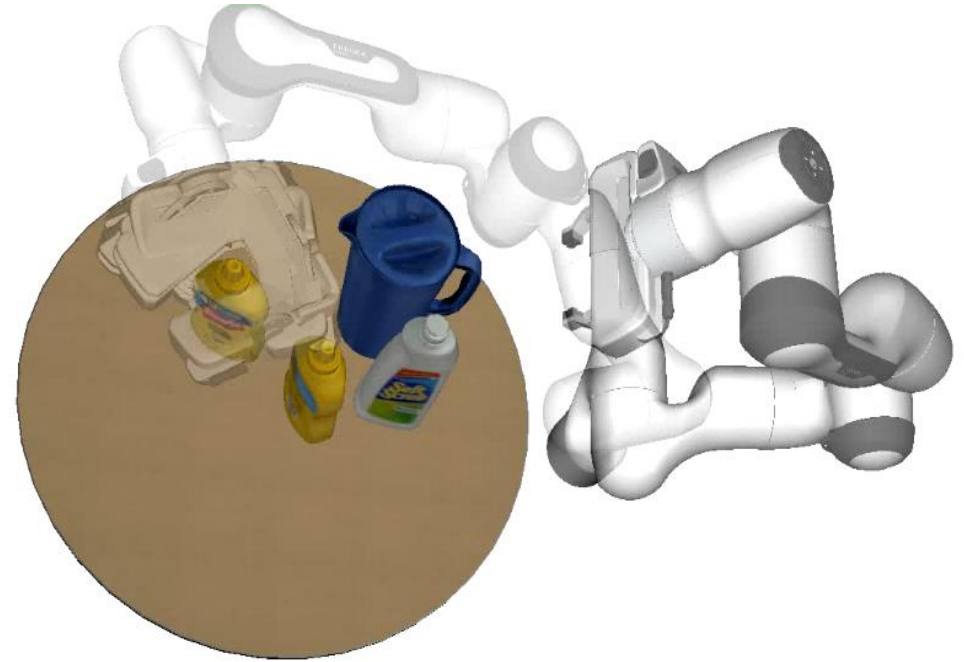


Motion Planning

Initial Trajectory



Trajectory Optimized by CHOMP



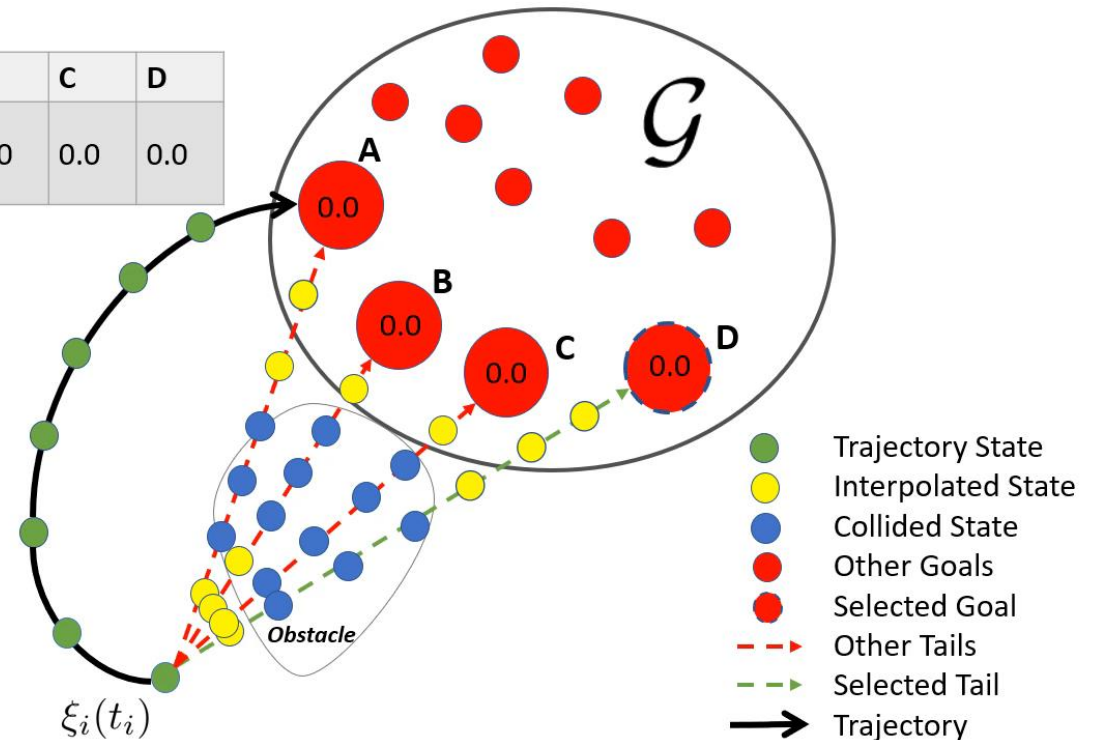
Online Learning for Grasp Selection

- Online Learning in a Bandit Setting
- Switch Goal during Motion Planning
- Simulated Cost-to-go

$$\mathcal{R}_N = \sum_{i=1}^N \langle c_i, p_i \rangle - \min_{g \in \mathcal{G}} \sum_{i=1}^N c_i(g)$$

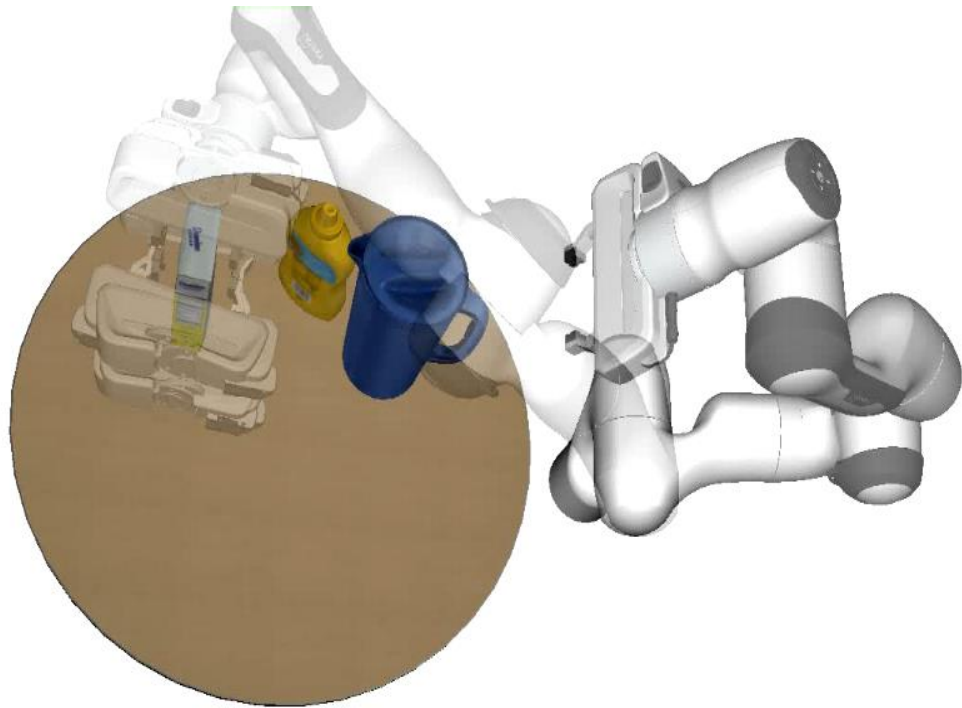
Iteration 0

Goal	A	B	C	D
Total Cost	0.0	0.0	0.0	0.0

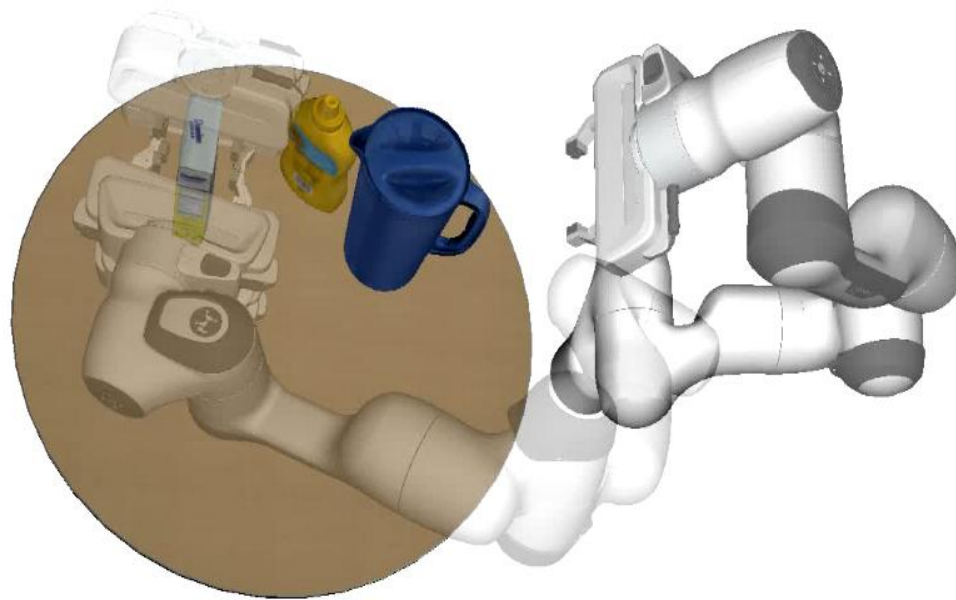


Goal Selection

Trajectory Optimized with CHOMP



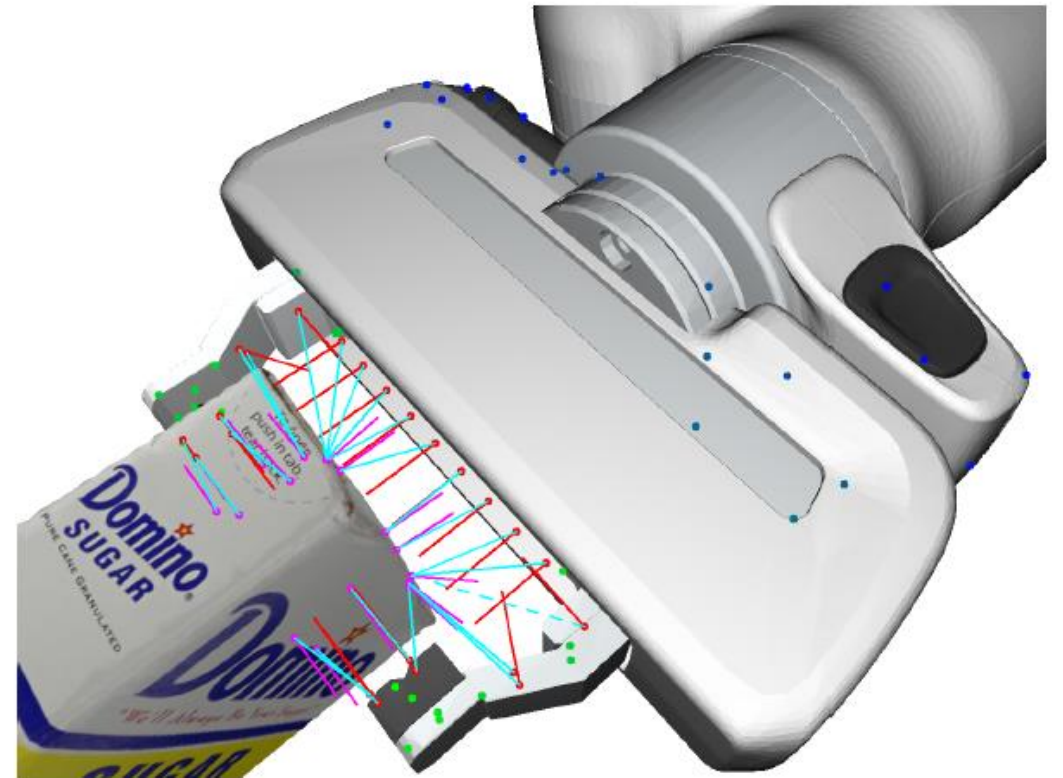
Trajectory Optimized with OMG



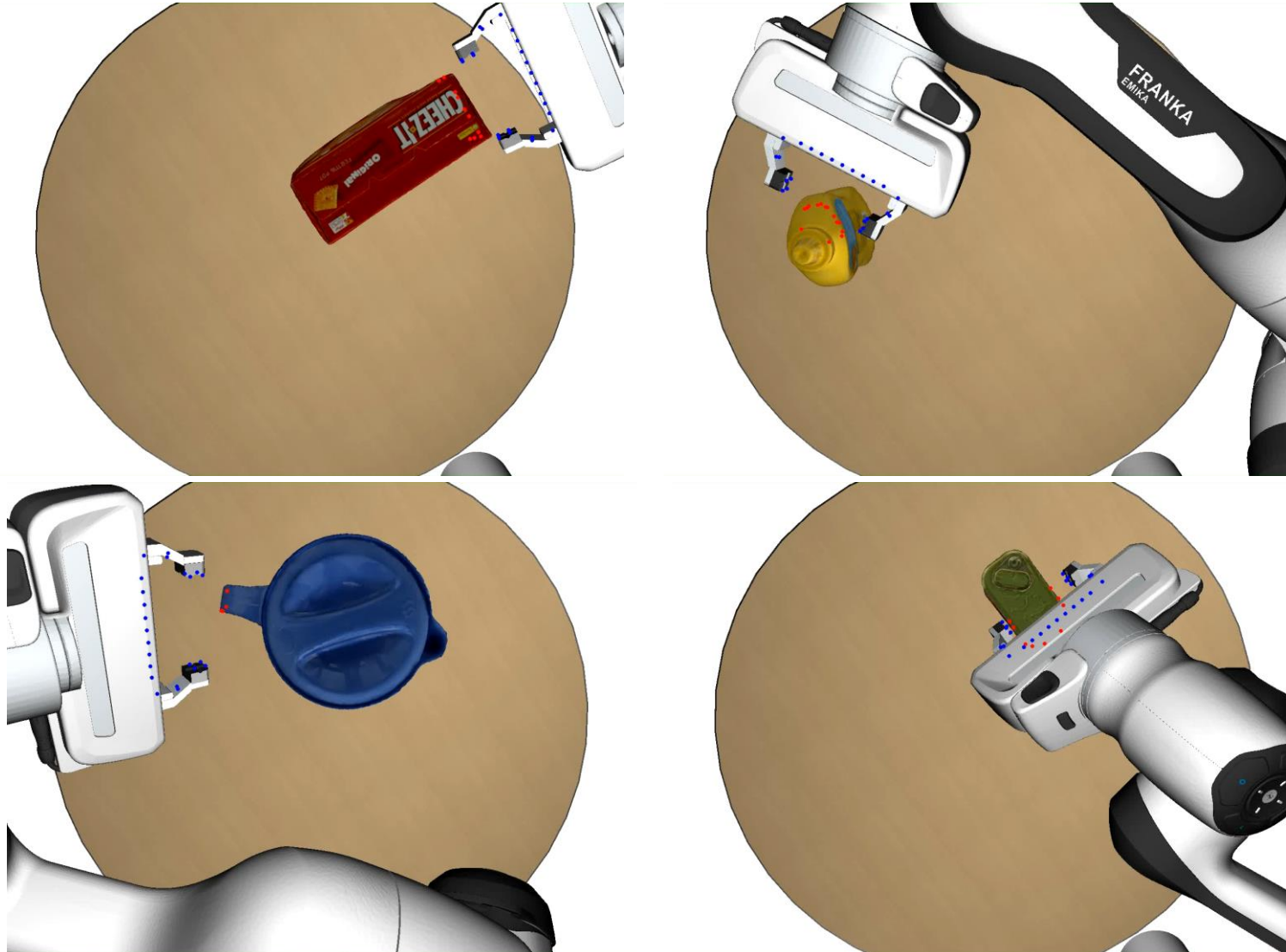
Iterative Surface Fitting for Grasp Refinement

$$f_{\text{grasp}}(g) = f_{\text{isf}}(g) + \gamma f_{\text{collision}}(g)$$

- Contact Point Distance and Normal Matching
- Penalize Hand Collision
- Pull Back via Exponential Map and Jacobian



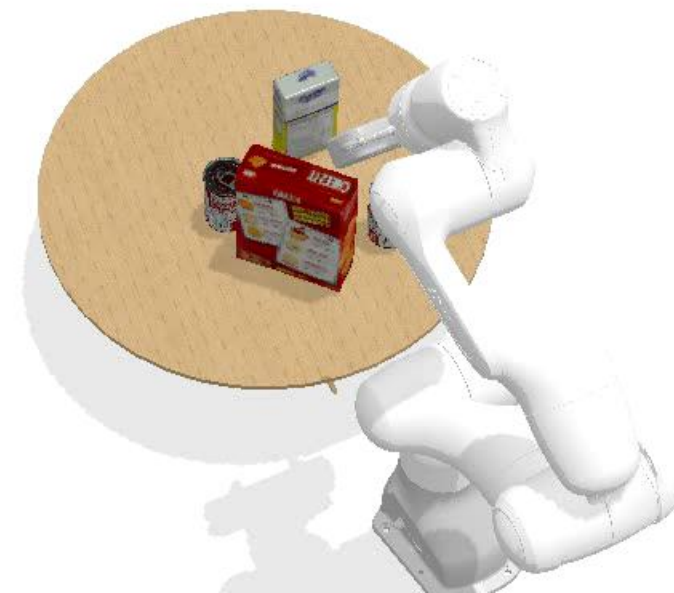
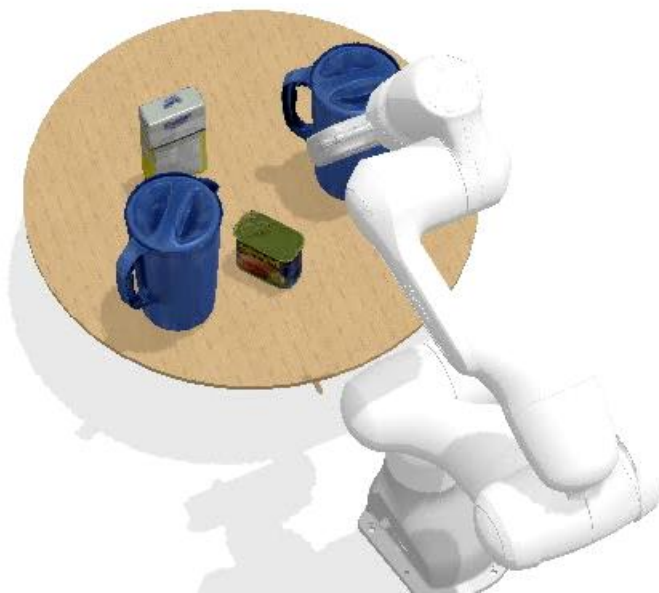
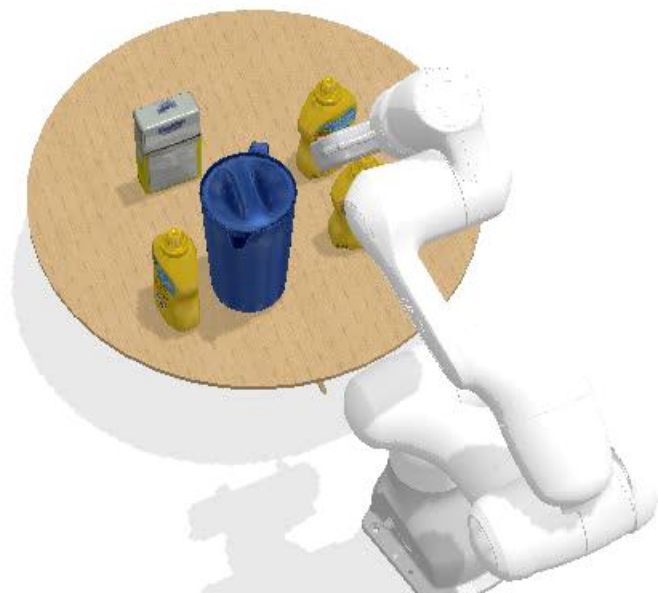
Goal Refinement

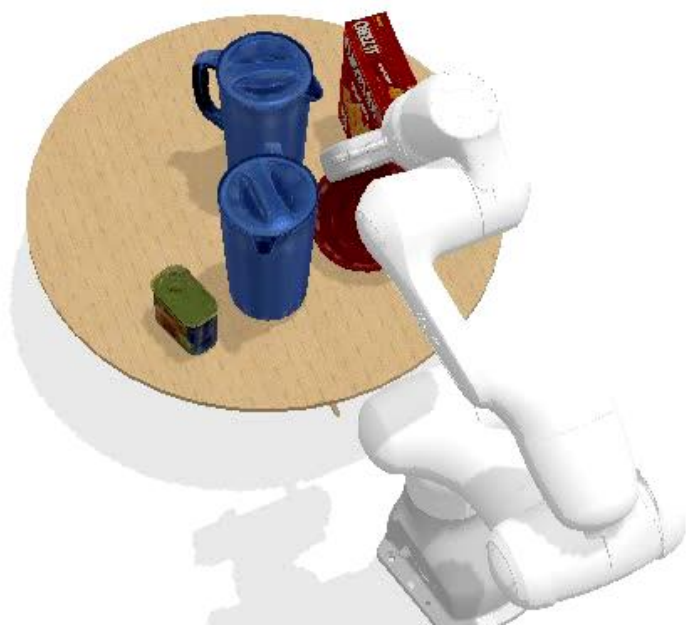


Comparison with Baseline

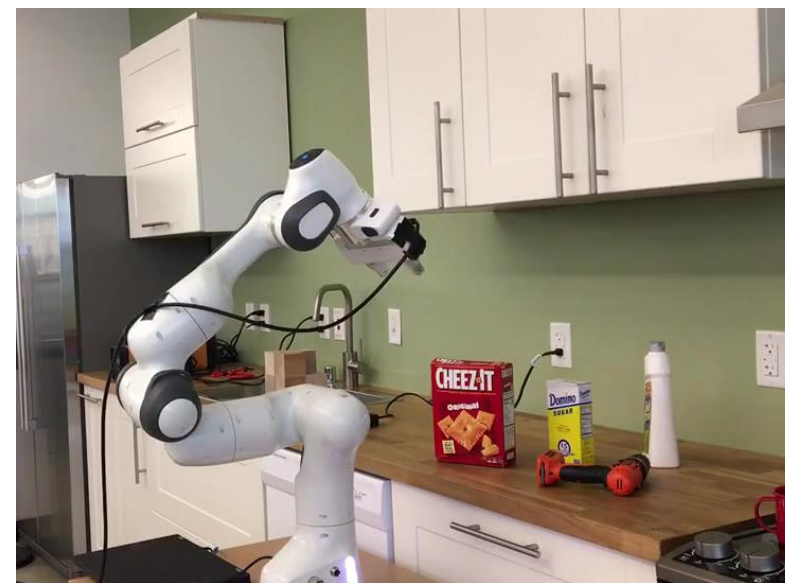
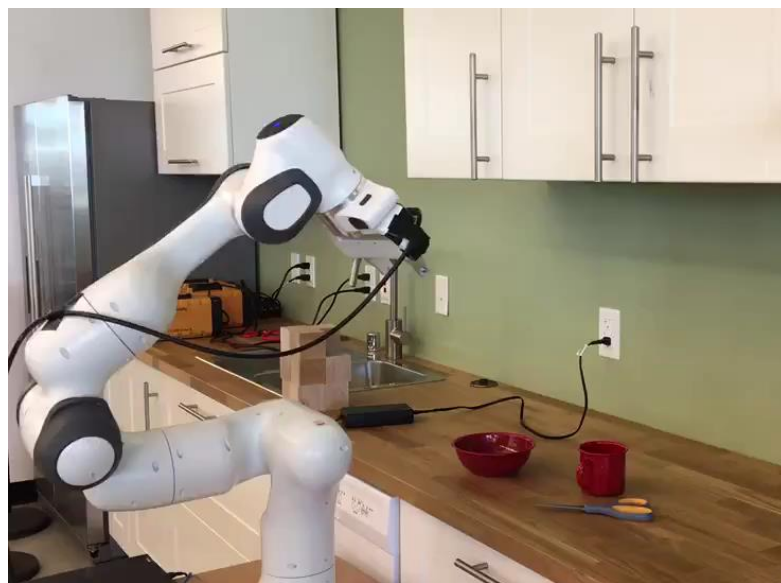
100 scenes, 30 grasps each scene, 10 runs					
Algorithm	CHOMP [33]	CHOMP-C [11]	FMT [21]	RRT-C [25]	OMG
Execution	77%	75%	66%	64%	84%
Planning	82%	79%	80%	78%	93%
Smoothness	19.12	11.24	27.21	26.45	15.43
Collision	4.78	12.06	6.37	6.52	3.92
Grasp	3.43	3.35	3.49	3.48	3.04
Time	0.25	0.15	0.67	0.22	0.42

Grasp Selection for Different Scenes



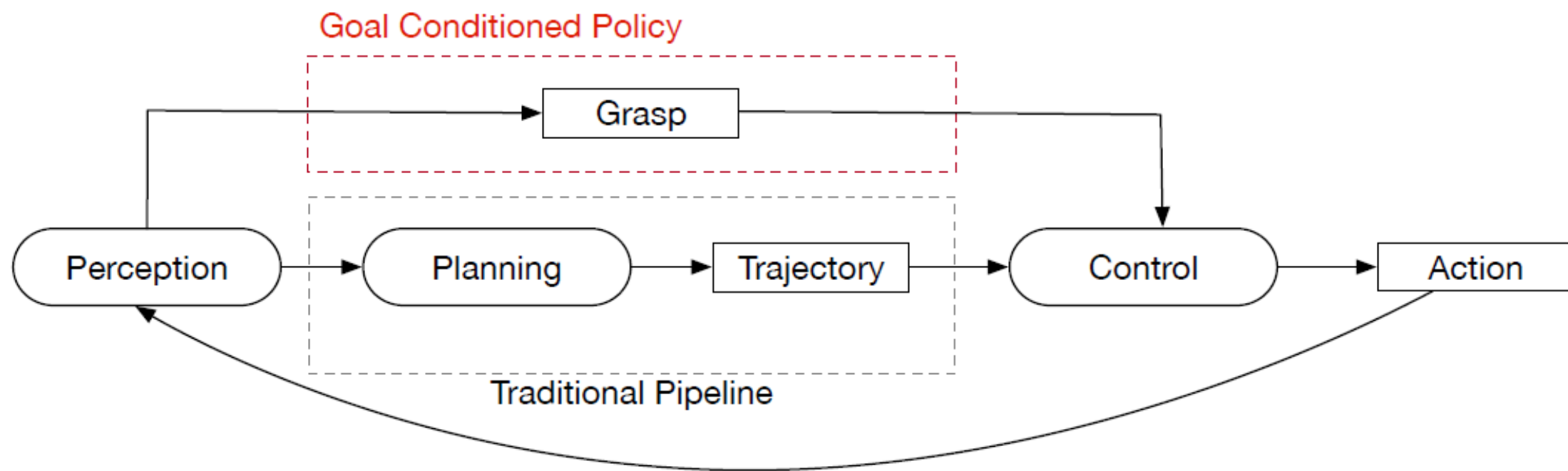


Real World Grasping





Control

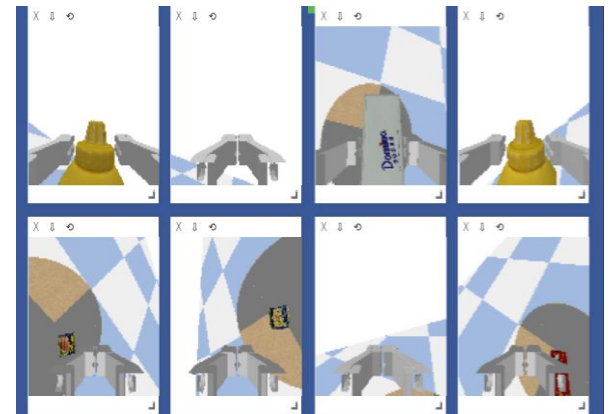
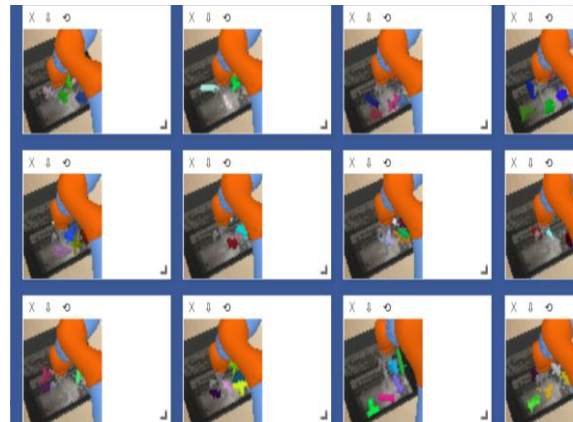


Grasping Policy

- Compared with Top-Down Grasping
 - Larger State Action Space
 - Harder Exploration
 - Complex Geometry and Contact

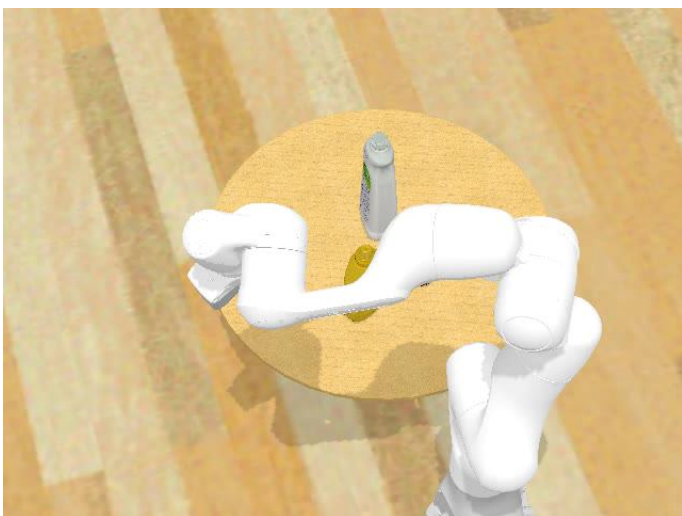
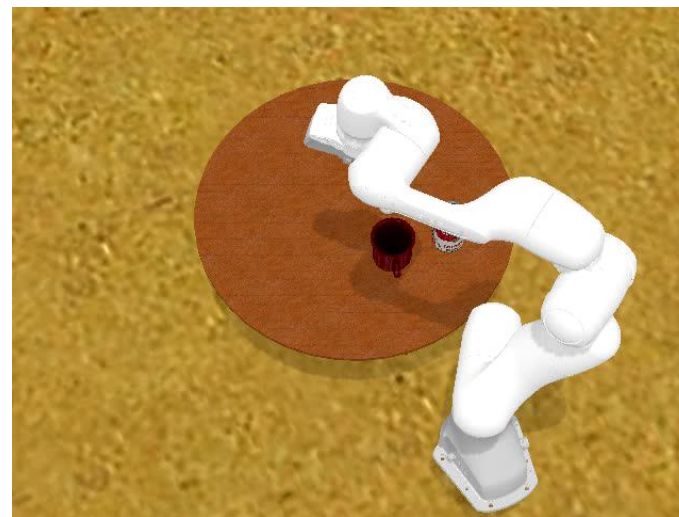


Real Env



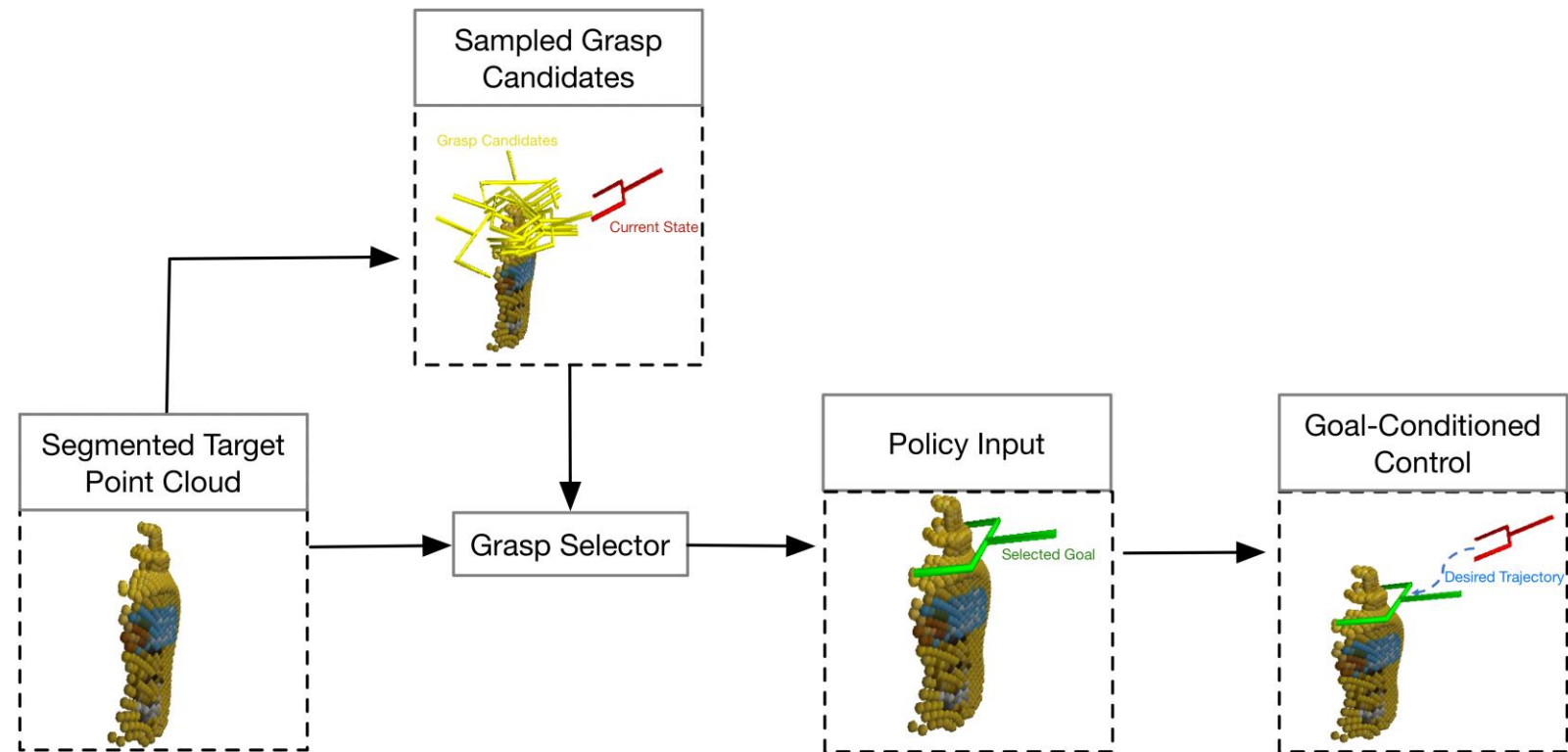
Simulation Env

Image-Based IL with Domain Randomization



Goal Conditioned Policy

- Observation: Target Point Cloud
- Feature Extractor: PointNet++
- Action: End-Effector Motion
- Reward: Sparse Binary Value
- Expert: Trajectory Planner



Goal-Conditioned Behavior Clone

- Goal-Conditioned Policy ($\pi_{\theta}(s, g)$)
- Subgoal Augmentation with Hindsight
- Goal-Reaching State Loss



Finetune with Deterministic Policy Gradient

- Deterministic Policy Gradient

$$\nabla_{\theta} \hat{J} = \mathbb{E}_{(s,a,g) \sim \mathcal{D}} [\nabla_a Q(s, a, g) \nabla_{\theta} \pi_{\theta}(s, g)]$$

- Off-Policy Actor Critic
- Sample efficient and work for continuous control
- Allow expert data and avoid IS over the action space
- Simple and straightforward to use with BC
- Improve the policy with on-policy rollouts

Grasping Trials



Preliminary Experiments

20 scenes per object, 5 runs			
Object Name	BC	GCG	GCG-Finetune
Cracker Box	0.56	0.97	0.95
Sugar Box	0.91	0.99	1.00
Tomato Soup Can	0.55	0.79	0.91
Mustard Bottle	0.71	0.65	0.78
Bleach Cleanser	0.83	0.92	0.94
Mean	0.712	0.864	0.916

Challenges and Future Work



- Challenges:
 - Complex Contact
 - Failure Recovery

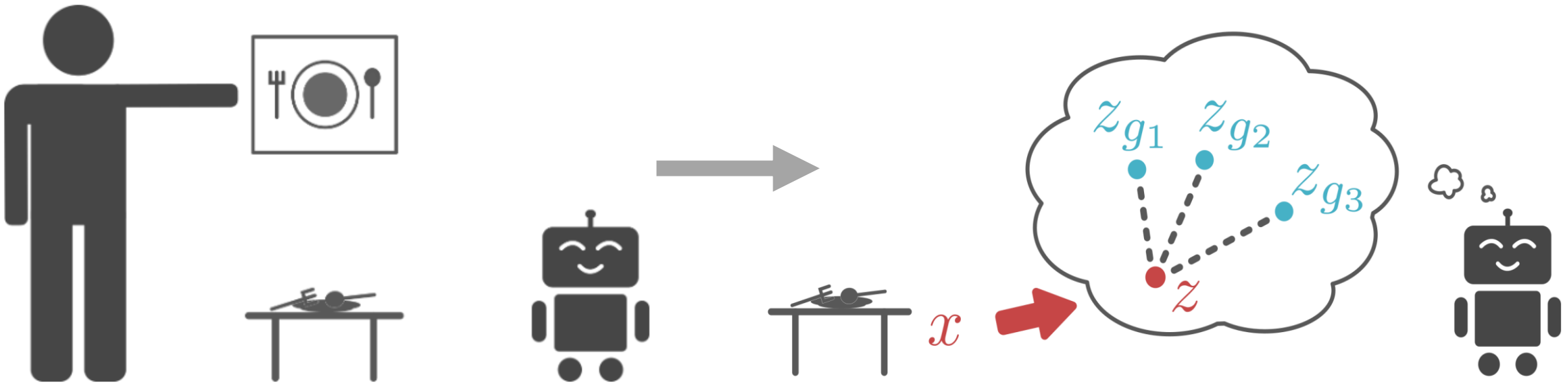


- Future Work:
 - More Experiment Objects
 - Combination of RL and IL
 - Goal-Conditioned Formulation

Conclusion

Man is a goal-seeking animal.

-- Aristotle



Thank you all!

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